



Exit quiz

© Oak National Academy 2024. Produced in partnership with Mathematics in Education and Industry.

Licensed on the [Open Government Licence v3.0](#), except where otherwise stated. [See Oak terms & conditions](#).



0

Extending thinking about sequences

1 In mathematics, sequences can be described as concrete or _____. Tick **1** correct answer



abstract



non-concrete

2 In 1980, the world marathon record was 2 hrs 10 mins. In the year 2000, it was 2 hrs 5 mins. In the year 2020, it was 2 hrs 0 mins. This is an example of _____ sequence. Tick **1** correct answer



a concrete



an abstract

3 In 1980, the world marathon record was 2 hrs 10 mins. In the year 2000, it was 2 hrs 5 mins. In the year 2020, it was 2 hrs 0 mins. Why is this concrete sequence limited? Tick **1** correct answer



It is not limited. The pattern will continue at -5 minutes per decade.



Records are decreasing now but can never reach zero. There are physical limits.



The record of 2hrs 0 mins cannot be broken.

4 What value would come before this geometric sequence? 196, 1372, 9604, 67 228, ... Fill in the blank

28

5 Scientists begin to study the population of a particular fish in the North Sea and find it forms a geometric sequence. Using the sequence to predict the population for year 5. Tick **1** correct answer

Year	1	2	3	4
Population	2 530 000	2 783 000	3 061 300	3 367 430



3 074 173



3 407 173



3 704 173



3 707 143

- 6 Scientists begin to study the population of a particular fish in the North Sea and find it forms a geometric sequence. Estimate the population of the fish the year before the study began. Fill in the blank

Year	1	2	3	4
Population	2 530 000	2 783 000	3 061 300	3 367 430

2300000